

Global AI & Data Jobs Salary Analysis

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Introduction

This project analyzes global AI and data job salaries from 2020–2026. The goal is to identify key factors that influence compensation and build predictive models. The data set was found using Kaggle and includes 90,000 observations across 35 variables, covering job roles, experience levels, locations, and work setups. The analysis will provide insights into salary trends and inform career strategies for professionals in the AI and data fields.

Research Questions

- What factors most influence salary?
- Do remote jobs pay more than onsite jobs?
- How do salaries vary by role, country, and experience?

Why This Matters

Understanding salary trends in AI and data jobs can help professionals make informed career decisions and guide employers in competitive compensation strategies. With the rapid growth of AI, insights from this analysis can also inform education and training programs to better prepare the workforce for future demands.

Data Overview

This data set includes variables like job role, country, experience level, salary in USD, and work setup (remote vs onsite). The data was collected from various job postings and salary surveys across different countries and industries. The main variables I want to focus on are salary, experience level, job role, and work setup, as these are likely to have the most significant impact on compensation.

```
import pandas as pd

df = pd.read_csv("global_ai_jobs.csv")

df.head()
```

	id	country	job_role	ai_specialization	experience_level	experience_yea
0	1	UAE	Machine Learning Engineer	Reinforcement Learning	Entry	0
1	2	USA	AI Engineer	LLM	Entry	1
2	3	Brazil	Research Scientist	Analytics	Entry	0
3	4	India	Software Engineer AI	Computer Vision	Senior	6
4	5	Germany	Machine Learning Engineer	Computer Vision	Entry	0

Data Cleaning

Before conducting analysis, the dataset was cleaned to ensure consistency, accuracy, and usability for statistical modeling.

Handling Missing Values

- First examined for missing values across all variables
- Columns with minimal missing values were handled by removing incomplete rows, while ensuring that a substantial portion of the data remained intact
- No columns with excessive missingness were retained, preserving the integrity of the dataset.

```
import pandas as pd

df = pd.read_csv("global_ai_jobs.csv")

df.columns = df.columns.str.lower().str.replace(" ", "_")
```

```
missing_values = df.isnull().sum()
missing_values
```

```
id                0
country           0
job_role          0
ai_specialization 0
experience_level  0
experience_years  0
salary_usd        0
bonus_usd         0
education_required 0
industry          0
company_size      0
interview_rounds  0
year             0
work_mode         0
weekly_hours      0
company_rating    0
job_openings      0
hiring_difficulty_score 0
layoff_risk       0
ai_adoption_score 0
company_funding_billion 0
economic_index    0
ai_maturity_years 0
offer_acceptance_rate 0
tax_rate_percent  0
vacation_days     0
skill_demand_score 0
automation_risk   0
job_security_score 0
career_growth_score 0
work_life_balance_score 0
promotion_speed   0
salary_percentile 0
cost_of_living_index 0
employee_satisfaction 0
dtype: int64
```

Data Type Corrections

Several variables were checked and converted into appropriate data types: - Salary was confirmed as a numeric variable - Categorical variables such as job role, country, experience level, and work setup were properly formatted - Any inconsistencies in formatting were corrected

```
df['salary_usd'] = pd.to_numeric(df['salary_usd'], errors='coerce')

# Drop rows with missing salary (critical variable)
df = df.dropna(subset=['salary_usd'])
```

Removing Duplicates

- Duplicate records were identified and removed to prevent bias in analysis and modeling
- This ensured that each job entry represented a unique observation
- Output shows 90,000 recorded observations after cleaning across 35 variables

```
df = df.drop_duplicates()
df.shape
```

(90000, 35)

After cleaning: - The dataset remained comprehensive and suitable for analysis - Key variables were consistent and properly formatted - The data was ready for exploratory analysis and modeling - Bases our insights off of reliable and well-constructed data

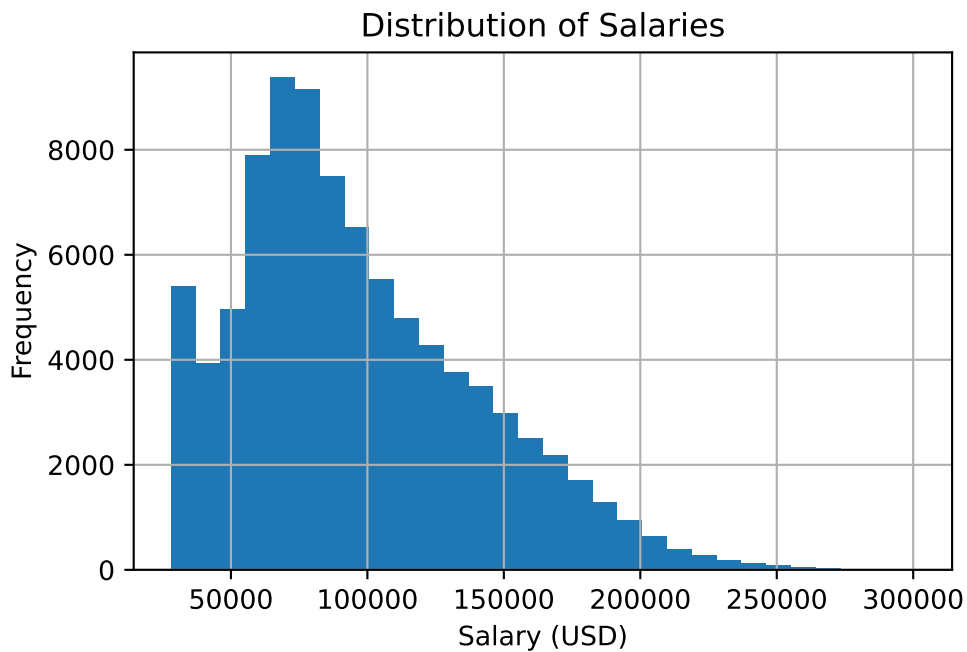
Exploratory Data Analysis

This section explores patterns in salaries across roles, experience levels, company sizes, and work setups to better understand the structure of the dataset and the behavior of the “salary” variable.

Salary Distribution

```
import matplotlib.pyplot as plt

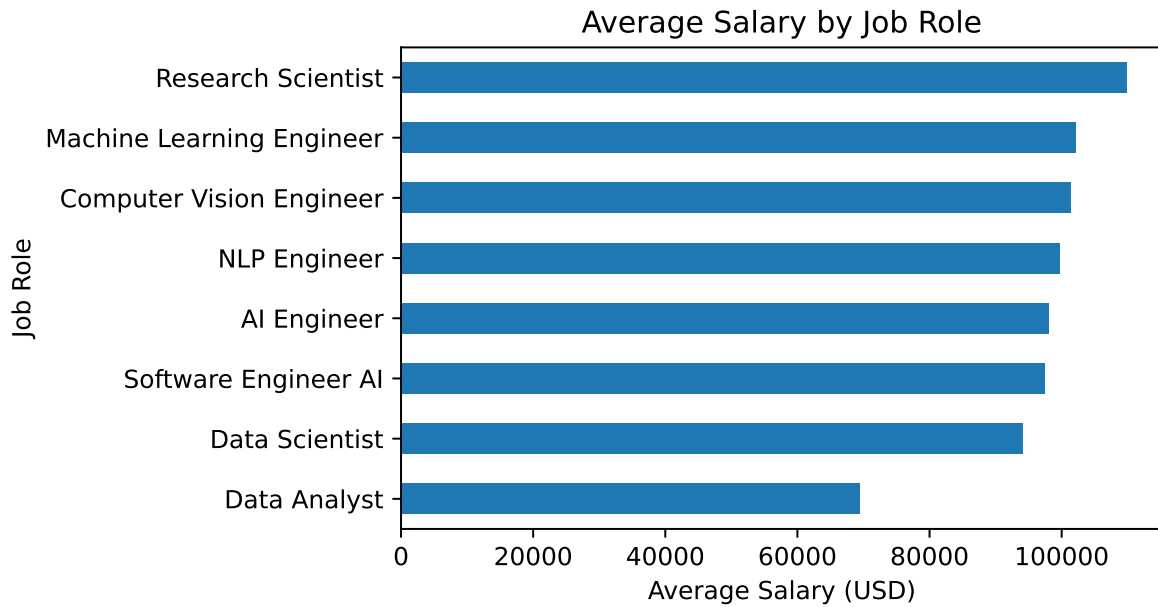
df['salary_usd'].hist(bins=30)
plt.title("Distribution of Salaries")
plt.xlabel("Salary (USD)")
plt.ylabel("Frequency")
plt.show()
```



- The salary distribution is right-skewed, indicating that while most jobs fall within a moderate salary range, a smaller number of roles offer significantly higher compensation
- These high-paying roles likely reflect senior or specialized positions.

Salary by Job Role

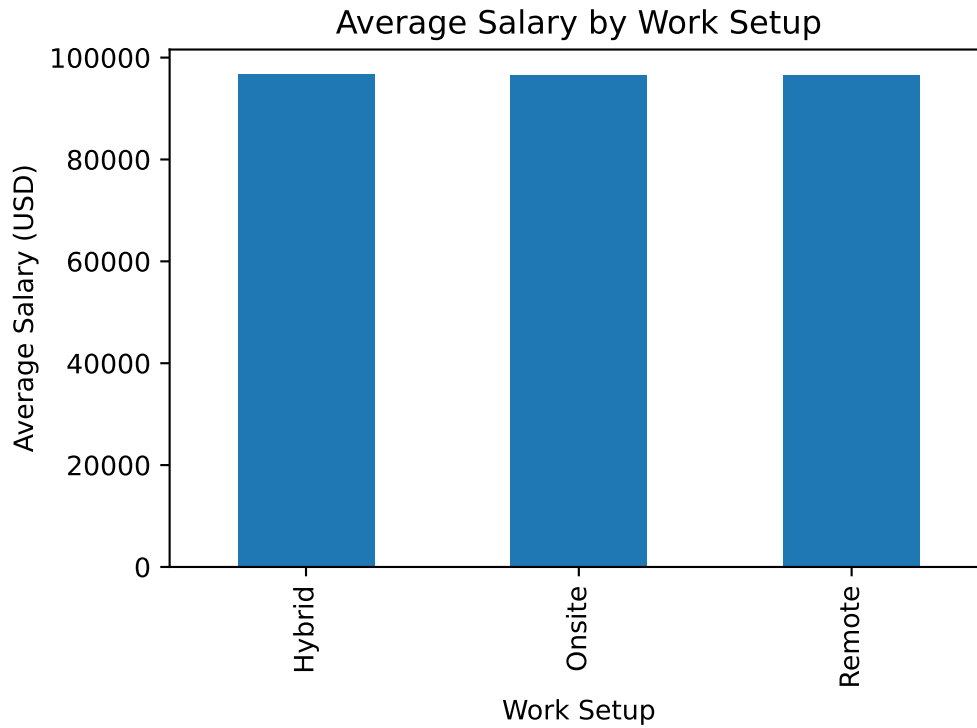
```
df.groupby('job_role')['salary_usd'].mean().sort_values().plot(kind='barh')
plt.title("Average Salary by Job Role")
plt.xlabel("Average Salary (USD)")
plt.ylabel("Job Role")
plt.show()
```



- There are clear differences in compensation across roles
- Machine learning and research scientists tend to command higher salaries compared to general analytics positions, reflecting higher demand and specialized skill requirements.

Remote vs Onsite Salaries

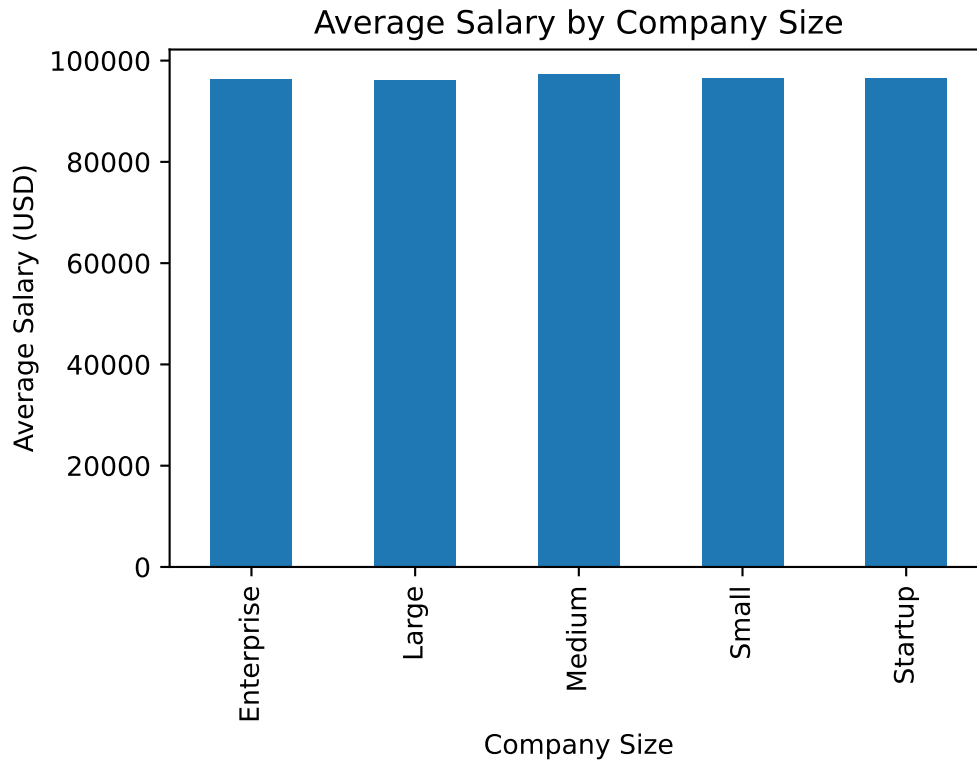
```
df.groupby('work_mode')['salary_usd'].mean().plot(kind='bar')
plt.title("Average Salary by Work Setup")
plt.xlabel("Work Setup")
plt.ylabel("Average Salary (USD)")
plt.show()
```



- Remote, hybrid, and onsite roles show minimal differences in average compensation
- This suggests that workplace flexibility may not be associated with salary, though further statistical testing is needed to confirm significance.

Salary by Company Size

```
df.groupby('company_size')['salary_usd'].mean().plot(kind='bar')
plt.title("Average Salary by Company Size")
plt.xlabel("Company Size")
plt.ylabel("Average Salary (USD)")
plt.show()
```



- Larger companies tend to offer similar salaries on average
- Indicates that organizational scale may not be a major factor in determining compensation

Hypothesis Testing

ANOVA Test for Salary Differences by Experience Level

H_0 : All experience levels have the same mean salary H_a : At least one experience level has a different mean salary

```
from scipy.stats import f_oneway

groups = [group['salary_usd'].values for name, group in df.groupby('experience_level')]

f_stat, p_value = f_oneway(*groups)

f_stat, p_value
```

```
(np.float64(28488.504051869677), np.float64(0.0))
```

The ANOVA test examining salary differences across experience levels produced an F-statistic of 28,488.50 and a p-value effectively equal to zero ($p < 0.001$). This result provides extremely strong evidence against the null hypothesis.

Therefore, we conclude that mean salaries differ significantly across experience levels. This finding confirms that experience is a major determinant of compensation in AI and data-related roles.

Remote vs Onsite Salary Comparison (Two-Sample t-test)

- H_0 : There is no difference in mean salary between remote and onsite jobs
- H_a : There is a significant difference in mean salary between remote and onsite jobs

```
from scipy.stats import ttest_ind

remote = df[df['work_mode'] == 'Remote']['salary_usd']
onsite = df[df['work_mode'] == 'Onsite']['salary_usd']

t_stat, p_value = ttest_ind(remote, onsite, nan_policy='omit')

t_stat, p_value
```

```
(np.float64(-0.022858829218200098), np.float64(0.9817629570474155))
```

The two-sample t-test comparing salaries between remote and onsite roles produced a t-statistic of -0.023 and a p-value of 0.982. Since the p-value is substantially greater than 0.05, we fail to reject the null hypothesis.

This indicates that there is no statistically significant difference in average salaries between remote and onsite positions. In other words, any observed difference in salaries is likely due to random variation rather than a true underlying effect.

Statistical Testing

Modeling and Results

To understand the drivers of salary and build predictive insight, two models were developed: a linear regression model and a random forest model.

Data Preparation for Modeling

Before modeling, categorical variables were encoded and the dataset was split into training and testing sets. - The target variable is salary in USD - Features include experience level, job role, country, company size, and work mode

```
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import OneHotEncoder
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline

y = df['salary_usd']

X = df[['experience_level', 'job_role', 'country', 'company_size', 'work_mode']]

# Identify categorical columns
categorical_cols = X.select_dtypes(include=['object', 'string']).columns
# Preprocessing: One-hot encoding
preprocessor = ColumnTransformer(
    transformers=[
        ('cat', OneHotEncoder(handle_unknown='ignore'), categorical_cols)
    ],
    remainder='passthrough'
)

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

Linear Regression Model

```
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

# Create pipeline
lr_model = Pipeline(steps=[
    ('preprocessor', preprocessor),
    ('model', LinearRegression())
])

# Train model
lr_model.fit(X_train, y_train)
```

```
# Predictions
y_pred_lr = lr_model.predict(X_test)

# Evaluation
import numpy as np
rmse_lr = np.sqrt(mean_squared_error(y_test, y_pred_lr))
r2_lr = r2_score(y_test, y_pred_lr)

rmse_lr, r2_lr
```

```
(np.float64(17202.333242434688), 0.8459599223534962)
```

Random Forest Model

```
from sklearn.ensemble import RandomForestRegressor

rf_model = Pipeline(steps=[
    ('preprocessor', preprocessor),
    ('model', RandomForestRegressor(n_estimators=100, random_state=42))
])

# Train model
rf_model.fit(X_train, y_train)

# Predictions
y_pred_rf = rf_model.predict(X_test)

# Evaluation
rmse_rf = np.sqrt(mean_squared_error(y_test, y_pred_rf))
r2_rf = r2_score(y_test, y_pred_rf)

rmse_rf, r2_rf
```

```
(np.float64(15121.706378670864), 0.880968853191937)
```

Model Performance and Interpretation

The linear regression model achieved an R^2 of 0.846, indicating that approximately 84.6% of the variation in salaries is explained by the model. This suggests a strong relationship between the

selected features and salary. The model's RMSE of 17,202 indicates that predictions deviate from actual salaries by about \$17,200 on average.

The random forest model further improved predictive performance, achieving an R^2 of 0.881. This means that 88.1% of the variation in salary is explained, reflecting a better overall fit. Additionally, the RMSE decreased to approximately \$15,122, indicating more accurate predictions compared to the linear regression model.

The improvement in performance suggests that salary relationships are not purely linear. The random forest model is able to capture nonlinear interactions between variables, such as how experience level and job role jointly influence compensation, leading to better accuracy.

Overall, both models perform well, but the random forest provides superior predictive capability. However, the linear regression model remains valuable for interpretation, as it clearly shows the direction and magnitude of relationships between individual factors and salary.

Key Insights

The analysis of global AI and data job salaries reveals several important patterns regarding compensation across roles, experience levels, and work environments.

Experience is the Strongest Driver of Salary

Across all analyses, experience level consistently emerged as the most significant factor influencing salary. Both the exploratory analysis and modeling results show that salaries increase substantially with experience, with senior and executive roles commanding significantly higher compensation. This highlights the importance of career progression in maximizing earnings within the AI and data fields.

Work Setup Has a Limited Impact

Differences between remote, hybrid, and onsite roles were observed, though their impact on salary is smaller compared to experience and job role. While remote work may offer flexibility, it does not appear to be the primary driver of compensation differences.

Company Characteristics Don't Influence Pay

Larger companies generally did not offer higher salaries. This suggests that organizational scale does not play a major role in compensation structures within the industry.

Nonlinear Relationships Improve Prediction Accuracy

The superior performance of the random forest model compared to linear regression indicates that salary is influenced by complex, nonlinear interactions between variables. Factors such as experience, role, and location do not operate independently but combine in ways that significantly affect compensation.

Overall Insight

Salary in AI and data-related roles is driven by a combination of experience, skills, and job specialization. While no single factor determines compensation, experience level and role specialization consistently emerge as the most influential drivers.

Recommendations

- For students and job seekers:
 - Prioritize gaining hands-on experience (internships, projects)
 - Develop specialized skills in AI and machine learning
 - Focus on high-demand technical areas to increase earning potential
- Career strategy:
 - Prioritize role specialization and growth opportunities over other variables
 - Remote work offers flexibility but does not significantly impact salary
- For employers:
 - Offer competitive salaries for experienced and specialized roles
 - Align compensation with market demand for high-skill talent
 - Recognize the value of retaining top technical employees

Limitations

- The dataset may not fully represent all global AI and data jobs, leading to potential sampling bias
- Salary data is reported in USD and may not account for differences in cost of living across countries

- Linear regression assumes linear relationships, which may oversimplify complex salary dynamics
- Random forest improves accuracy but reduces interpretability of individual variable effects
- External factors such as company performance, industry trends, and negotiation are not captured

Future Work

- Incorporate additional variables such as education level, certifications, and years of experience for deeper analysis
- Adjust salaries for cost of living to better compare compensation across countries
- Conduct longitudinal analysis to better understand salary trends over time
- Expand the dataset to include more recent data and broader industry coverage

Thank you, questions?

Sources

Thalla, Mohan Krishna. “Global AI & Data Jobs Salary Dataset.” Kaggle, 2026, <https://www.kaggle.com/datasets/mohankrishnathalla/global-ai-and-data-jobs-salary-dataset>. Galarita, Brandon. “Data Science Salary: How Much Can You Make With a Data Science Degree?” Forbes Advisor, 2024, <https://www.forbes.com/advisor/education/it-and-tech/data-science-salary/>.